

Breathing and Exchange of Gases

1. Read the following statements w.r.t. humans.

- (a) Exchange part of the respiratory system is devoid of incomplete cartilaginous rings.
- (b) Maximum O₂ is carried by haemoglobin.
- (c) Binding of O₂ with Hb primarily depends upon the levels of pCO₂.

Choose the option which represents the incorrect statement(s) only.

- (1) (a) and (b)
- (2) (c)
- (3) (b)
- (4) (b) and (c)

2. Given below are two statements: One is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion : Rhythmic centre present in the medulla region of the brain.

Reason : Neural signal from medulla centre can reduce the duration of inspiration and thereby alter the respiratory rate.

Choose the most appropriate answer

- (1) Assertion and reason both are true and the reason is correct explanation of assertion.
- (2) Assertion and reason both are true and the reason is not correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are wrong.

3. Which of the following statement is correct.

- (1) Residual volume is not measured by spirometer
- (2) Lungs are covered by pericardium
- (3) tidal volume is 1000 ml in healthy human
- (4) functional and structural unit of lung is trachea

4. Among vertebrates, fishes use _____ whereas reptiles and birds respire through _____.

- (1) skin, lungs
- (2) gills, cuticle
- (3) gills, lungs
- (4) gills, spiracles

5. Chronic cigarette smoking leads to a chronic respiratory disorder that is characterized by

- (1) Decreased respiratory surface
- (2) Allergic inflammation of the mast cells
- (3) Wheezing due to inflammation of alveoli
- (4) Inflammation of bronchioles only

6. In humans, the major part of carbon dioxide is transported

- (1) As bicarbonate
- (2) As carboxyhaemoglobin
- (3) As carbaminohaemoglobin
- (4) In a dissolved state through blood plasma

7. Match List-I with List-II

List – I

List - II

- I. Earthworm
- II. Aquatic arthropods
- III. Cockroach
- IV. Birds and reptiles

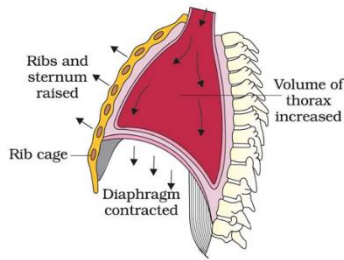
- A. Moist cuticle
- B. Gills
- C. Lungs
- D. Tracheal tubes

- (1) I-B, II-A, III-D, IV-C
- (2) I-A, II-D, III-C, IV-B
- (3) I-A, II-B, III-D, IV-C
- (4) I-A, II-B, III-C, IV-D

8. Oxygen dissociation curve is.

- (1) S-shaped
- (2) M-shaped
- (3) Z-shaped
- (4) L-shaped

9. Which of the following option indicated the breathing stage of the diagram



- (1) Expiration
- (2) Inspiration
- (3) Exhalation
- (4) Both 1 and 3

10. Which is the correct sequence of air passage in man?

- (1) External nostrils → Nasal passage → Internal nares → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli
- (2) Nose → Larynx → Pharynx → Bronchioles → Bronchi → Alveoli
- (3) Nose → Pharynx → Larynx → Bronchioles → Bronchi → Alveoli
- (4) Nose → Bronchioles → Pharynx → Larynx → Bronchi → Alveoli

11. Consider the given statements and select the correct option.

Statement I: Reduction in pH of blood increases the affinity of haemoglobin with oxygen.

Statement II: Silicosis and pneumonia are the examples of occupational respiratory disorders.

- (1) Both statements I and II are correct.
- (2) Both statements I and II are incorrect.
- (3) Only statement I is correct.
- (4) Only statement II is correct.

12. Megakaryocytes are the special cells in the bone marrow that upon fragmentation give rise to

- (1) Enucleated cell fragments
- (2) Nucleated cell fragments
- (3) Clotting factors
- (4) Nucleated thrombocytes

13. Match the column-I with column-II

	Column-I		Column-II
I.	Larynx	A.	Straight tube extending through mid thoracic cavity
II.	Epiglottis	B.	Common passage for food and air
III.	Pharynx	C.	Covers glottis during swallowing
IV.	Trachea	D.	Sound box

- (1) I-B, II-A, III-D, IV-C
- (2) I-A, II-B, III-C, IV-D
- (3) I-D, II-C, III-B, IV-A
- (4) I-A, II-B, III-D, IV-C

14. Difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles. These symptoms indicate.

- (1) Asthma
- (2) Emphysema
- (3) Occupational respiratory disorders
- (4) None of these

15. On an average, a healthy human breathes _____ times/minute under normal physiological conditions.

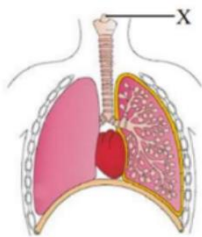
Choose the option that fills the blank correctly.

- (1) 15-20
- (2) 12-16
- (3) 10-12
- (4) 20-25

16. Which of the following is not the function of the conducting part of respiratory system in human?

- (1) Trapping foreign particles
- (2) Humidifying the inspired air
- (3) Exchange of O₂ and CO₂ between blood and atmospheric air
- (4) Adjustment of air temperature to body temperature

17. Which what is the function of structure marked as X ?



- (1) To prevent the entry of food into larynx.
- (2) To filter and warm the air.
- (3) To help in exchange of gases.
- (4) To catch dust and bacteria.

18. Given below are two statements:

Assertion (A): The lungs are situated in the thoracic chamber which is anatomically an air tight chamber.

Reason (R): The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the lung (pulmonary) cavity.

In the light of the above statements, choose the most appropriate answer from option given below

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, but Reason (R) is false.
- (4) Assertion (A) is false, but Reason (R) is true.

19. Approximately 70% of carbon di-oxide absorbed by the blood will be transported of the lungs.

- (1) as bicarbonate ions
- (2) in the form of dissolved gas molecules
- (3) by binding to RBC
- (4) as carbamino-haemoglobin

20. Complete the analogy by choosing the correct option w.r.t. the respiratory structures of animals. Molluscs : Gills :: Flatworms _____

- (1) Tracheal tubes
- (2) Body surface
- (3) Gills
- (4) Lungs

21. Match the column-I to column-II and select the correct option.

Column-I	Column-II
(a) Inspiratory capacity	(i) IRV + EC
(b) Vital capacity	(ii) TV + IRV
(c) Residual volume	(iii) TLC – VC
(d) FRC	(iv) TLC – IC

- (1) a–i; b–iii; c–ii; d–iv
- (2) a–iv; b–i; c–iii; d–i
- (3) a–iii; b–i; c–ii; d–iv
- (4) a–ii; b–i; c–iii; d–iv

22. Which of the following is the function of pneumocytes?

- (1) It helps in gaseous exchange.
- (2) Secretion of lecithin, which act as a surfactant.
- (3) Prevention of alveolar collapse.
- (4) All of these

23. In humans, for regulation of respiration, a specialised centre is present in the medulla region of brain called 'X' and another specialised centre called 'Y' is present in pons. Choose the correct option w.r.t. 'X' and 'Y'.

- (1) 'X' can moderate the functions of 'Y', which further decreases the respiratory rate
- (2) Neural signals from 'Y' can influence 'X' and can lead to increase in the respiratory rate
- (3) A chemosensitive area is present adjacent to 'Y'
- (4) An increase in concentration of O₂ and pH immediately activates 'X' to make necessary adjustments in the respiratory process

24. Given below are two statements;

Statement-I : Inspiration is initiated by the contraction of diaphragm which increases the volume of thoracic chamber in the dorso ventral axis.

Statement-II: The contraction of external inter-costal muscles lifts up the ribs and the sternum causing an increase in the volume of the thoracic chamber in the antero-posterior axis.

Choose the most appropriate answer from the options given below :

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

25. Which of the following does not occur during breathing.

- (1) Brings the air to body temperature
- (2) Diffusion of gases
- (3) Warms up the air
- (4) Cleans up the air

26. Match the column-I with Column-II and select the correct option.

Column-I	Column-II
(a) Tidal volume	(i) 2500-3000 mL
(b) IRV	(ii) 500 mL
(c) ERV	(iii) 1100-1200 mL
(d) Residual volume	(iv) 1000-1100 mL

Choose the most appropriate answer from the options given below :

- (1) a-ii; b-i; c-iv; d-iii
- (2) a-i; b-ii; c-iii; d-iv
- (3) a-ii; b-i; c-iii; d-iv
- (4) a-iv; b-i; c-iii; d-ii

27. During normal inspiration in humans, the

- (a) Ribs and sternum lift up and forward
- (b) Thoracic volume reduces
- (c) Intra-pulmonary pressure decreases than that of atmospheric pressure

Choose the correct option.

- (1) (a) and (b)
- (2) (a) and (c)
- (3) Only (b)
- (4) (a), (b) and (c)

28. In humans, under normal physiological conditions, the value of $p\text{CO}_2$ in alveoli is

- (1) Equal to that of $p\text{CO}_2$ in tissues
- (2) More than that of $p\text{O}_2$ in aorta
- (3) Equal to that of $p\text{O}_2$ in tissues
- (4) Less than that of $p\text{CO}_2$ in atmospheric air

29. The volume of air that remains in the lungs after a normal expiration in healthy adult humans is

- (1) 2100 – 2300 mL
- (2) 1100 – 1200 mL
- (3) 2500 – 3000 mL
- (4) 1000 – 1100 mL

30. Arrange the following in the order of increasing volume.

I. Tidal volume

II. Residual volume

III. Expiratory reserve volume

IV. Vital capacity

- (1) I < II < III < IV
- (2) I < III < II < IV
- (3) I < IV < III < II
- (4) I < IV < II < III

31. How much CO_2 is delivered to the alveoli by every 100 ml of deoxygenated blood.

- (1) 6 ml
- (2) 4 ml
- (3) 5 ml
- (4) 3 ml

32. Given below are two statements:

Statement-I : Neural signals from the pneumotaxic centre can reduce the duration of inspiration and thus, alter the respiratory rate.

Statement-II: Occupational respiratory disorder can result in the proliferation of fibrous tissue in lungs.

Choose the most appropriate answer

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

33. All of the following respiratory volumes/capacities can be estimated by a simple spirometer, except

- (1) Tidal volume
- (2) Expiratory capacity
- (3) Total lung capacity
- (4) Inspiratory reserve volume

34. In humans, trachea divides at the level of 5 th thoracic vertebra into

- (1) Primary bronchi
- (2) Secondary bronchi
- (3) Tertiary bronchi
- (4) Terminal bronchiole

35. Due to increasing air-borne allergens and pollutants, many people in urban areas are suffering from respiratory disorder that cause wheezing due to;

- (1) reduction in the secretion of surfactant by pneumocytes.
- (2) benign growth on mucous lining of nasal cavity.
- (3) inflammation of bronchi and bronchioles.
- (4) proliferation of fibrous tissues and damage of the alveolar walls.

36. Given below are two statements:

Assertion (A): When CO₂ concentration in blood increases, breathing becomes slower and deeper.

Reason (R): The CO₂ released in respiring tissue accelerates the binding of oxygen to the haemoglobin by means of slow and deep breathing.

In the light of above statements, select the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

37. Solubility of CO₂ in the blood is ____ times higher than O₂.

Fill the blank with a suitable option.

- (1) 120-125
- (2) 80-100
- (3) 20-25
- (4) 40-70

38. Given below are some factors that affect the oxygen dissociation curve. Which of the following factors shift the curve towards the left side?

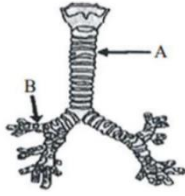
- I. Low partial pressure of O₂
- II. High temperature
- III. Low hydrogen ion concentration
- IV. Low partial pressure of CO₂

- (1) I and II
- (2) III and IV
- (3) I and IV
- (4) II and III

39. Transport of gases in alveoli takes place by.

- (1) Active transport
- (2) Passive transport
- (3) Simple diffusion
- (4) None of these

40. In the given figure, label A represents _____ while label B represents _____.



- (1) A-Trachea, B-Bronchus
- (2) A-Alveolus, B-Bronchiole
- (3) A-Bronchiole, B-Trachea
- (4) A-Trachea, B-Bronchiole

41. Given below are two statements:

Assertion (A): Volume of fresh air which take part in gaseous exchange per minute is called alveolar ventilation rate.

Reason (R): Residual volume can not be measured by spirometer because it is difficult to breath out completely.

Choose the most appropriate answer

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, but Reason (R) is false.
- (4) Assertion (A) is false, but Reason (R) is true.

42. By the contraction of diaphragm, thoracic volume is increased in.

- (1) dorso ventral axis
- (2) antero posterior axis
- (3) dorso lateral axis
- (4) antero lateral axis

43. The factor which does not affect the rate of alveolar diffusion is;

- (1) Solubility of gases
- (2) Thickness of the membranes
- (3) Pressure gradient
- (4) Reactivity of the gases

44. Which of the following statement is not true?

- (1) The partial pressure of oxygen in deoxygenated blood is 40 mm Hg.
- (2) The partial pressure of oxygen in oxygenated blood is 95 mm Hg.
- (3) The partial pressure of oxygen in the alveolar air is 104 mm Hg.
- (4) The partial pressure of carbon dioxide in deoxygenated blood is 40 mm Hg.

45. Given below are two statements:

Statement-I : At alveoli level, gaseous exchange occurs through respiratory membrane.

Statement-II: Gases move from the areas of higher partial pressure to areas of lower partial pressure.

Choose the most appropriate answer

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

46. Select the correct events that occur during inspiration

- (a) Contraction of diaphragm
- (b) Contraction of EICM
- (c) Pulmonary volume decreases
- (d) Intra pulmonary pressure increases

- (1) c and d
- (2) a, b and d
- (3) only d
- (4) a and b

47. Lungs are covered by

- (1) pericardium
- (2) Glisson's capsule
- (3) pleural membrane
- (4) Renal capsule

48. Given below are two statements:

Assertion (A): CO₂ diffuses in RBCs and dissolve in water of cytoplasm to form carbonic acid.

Reason (R): Formation of carbonic acid occurs in the presence of carbonic anhydrase enzyme.

Choose the most appropriate answer

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, but Reason (R) is false.
- (4) Assertion (A) is false, but Reason (R) is true.

49. Which of the following is an occupational respiratory disease?

- (1) Diptheria
- (2) Pneumonia
- (3) Tuberculosis
- (4) Silicosis & Abestosis

50. Given below are two statements:

Statement-I : Haemoglobin is a red coloured iron containing pigment present in the RBCs.

Statement-II: O₂ can bind with haemoglobin in a irreversible manner to form oxyhaemoglobin.

Choose the most appropriate answer

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

Answer Key

- 1. (2)
- 2. (3)
- 3. (1)
- 4. (3)
- 5. (1)
- 6. (1)
- 7. (3)
- 8. (1)
- 9. (2)
- 10. (1)
- 11. (2)
- 12. (1)
- 13. (3)
- 14. (1)
- 15. (2)
- 16. (3)

17. (1)
18. (1)
19. (1)
20. (2)
21. (4)
22. (4)
23. (2)
24. (4)
25. (2)
26. (1)
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38. (2)
39. (3)
40. (1)
41. (2)
42. (2)
43. (4)
44. (4)
45. (1)
46. (4)
47. (3)
48. (2)
49. (4)
50. (2)

